Example Complete stack

```
mern-demo/
├── mern-backend/          # Express + MongoDB (server side)
│   ├── server.js          # Your backend entry file
│   ├── package.json       # Created by npm init
│   └── node_modules/      # Installed dependencies (express, mongoose, cors)
│
├── mern-frontend/         # React app (client side)
│   ├── public/            # Static files (index.html, etc.)
│   ├── src/
│   │   ├── App.js         # Your React code
│   │   └── index.js       # React entry point
│   ├── package.json       # Created by npx create-react-app
│   └── node_modules/      # Installed React dependencies
```

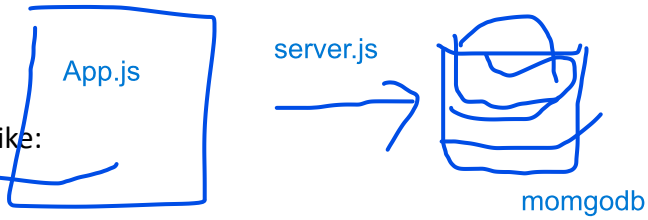
npm start 3000
node server.js 5000

npx create-react-app .

Server.js BACKEND

```
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');

const app = express();
app.use(cors());

// Connect to MongoDB
mongoose.connect('mongodb://localhost:27017/studentdb')
  .then(() => console.log('MongoDB Connected'))
  .catch(err => console.error(err));

// Define Student schema
const studentSchema = new mongoose.Schema({
  rollno: Number,
  name: String,
  course: String,
  semester: Number
});

const Student = mongoose.model('Student', studentSchema);

// Sample route: get all students
app.get('/students', async (req, res) => {
  const students = await Student.find();
  res.json(students);
});

// Start server
app.listen(5000, () => console.log('Server running on http://localhost:5000'));
```

MongoDB stores student data.
Express provides an API endpoint /students.
http://localhost:5000/students returns JSON like:




```
[
  { "rollno": 1, "name": "Amit", "course": "BSc CS", "semester": 3 },
  { "rollno": 2, "name": "Priya", "course": "MBA", "semester": 1 }
]
```

App.js Frontend

```
import React, { useEffect, useState } from 'react';

function App() {
 const [students, setStudents] = useState([]);

 useEffect(() => {
  fetch('http://localhost:5000/students')
   .then(res => res.json())
   .then(data => setStudents(data));
 }, []);

 return (
  <div>
   <h1>Student List</h1>
   <ul>
    {students.map(s => (
     <li key={s.rollno}>
      {s.rollno} - {s.name} ({s.course}, Semester {s.semester})
     </li>
    ))}
   </ul>
  </div>
 );
}

export default App;
```

BACKEND setup and Testing

Setup: Backend Setup (Express + MongoDB)

1. Create a folder for backend, e.g. mern-backend.
2. Open it in VS Code.

npm init -y
npm install express mongoose cors

project backend structure is created

start the backend server
node server.js

paste this server.js

Testing: Run MongoDB and insert student records.

```
db.students.insertMany([
  { rollno: 1, name: "Amit", course: "BSc CS", semester: 3 },
  { rollno: 2, name: "Priya", course: "MBA", semester: 1 }
])
```

Start Express (node server.js).
Start React (npm start).
Open http://localhost:3000 → Students list appears.

FRONT END Setup

Create mern-frontend folder
Execute command in this folder

npx create-react-app . **(dot indicates current directory)**

change to mern-frontend

replace code of App.js /src/App.js

npm start

Two separate projects:

- `mern-backend` → runs on Node.js with Express, talks to MongoDB.
- `mern-frontend` → runs on React, fetches data from backend.

Start commands:

- Backend: `node server.js` (runs on port 5000).
- Frontend: `npm start` (runs on port 3000).

Connection: React (`http://localhost:3000`) fetches data from Express (`http://localhost:5000/students`).